Data Sheet

EXPERIMENT NO. 05

Title: - Study of R-C Coupled Amplifier — Frequency Response and Bandwidth (using Tinkercad/ Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Voltage Gain: -

Gain (A) = Vo / Vi

- V_o = Output voltage
- V_i = Input voltage

Gain in Decibels: -

Gain (dB) = 20 log10 (Vo / Vi)

Bandwidth: -

Bandwidth (BW) = $f_H - f_L$

- f_H = Upper cut-off frequency (frequency at which gain falls to 0.707 of maximum / midband gain, i.e. −3 dB point)
- f_L = Lower cut-off frequency (frequency at which gain falls to 0.707 of maximum / midband gain, i.e. −3 dB point)

Recorded Parameters: -

Input Voltage, V_i = _____

Midband Voltage Gain = _____

Midband Voltage Gain in dB = _____

Lower Cut-off Frequency, f_L = _____

Upper Cut-off Frequency, f_H = _____

Bandwidth, BW = $f_H - f_L$ = _____

Observation Table: -

Sl. No.	Frequency (f) Hz	Output Voltage, Vo (V)	Log10 f	Gain = Vo/Vi	Gain (dB) = 20 log10(Vo/Vi)

Calculation: -

Midband Gain (A) = Vo / Vi = _____

Midband Gain in dB = 20 log10 (A) = _____

Bandwidth (BW) = $f_H - f_L$ = _____

Conclusion: -

Teacher's Sign and Date